- Scalar matrix: $A = kI$
- $Ae_j \rightarrow j^{th}$ column of A
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(AB)^t = B^tA^t$, $(AB)^* = B^*A^*$ $\quad A^* \rightarrow$ adjoint of A
- $(A^{-1})^t = (A^t)^{-1}$
- $E[i,j] \rightarrow i^{th}$ and j^{th} rows exchange

  $E_\alpha[i] \rightarrow i^{th}$ row multiplied with non-zero 'α'

  $E_\alpha[i,j] \rightarrow i^{th}$ row replaced by i^{th} row + $\alpha \times j^{th}$ row
- $E[i,j]^{-1} = E[i,j]$ $\quad E_\alpha[i]^{-1} = E_{1/\alpha}[i]$ $\quad E_\alpha[i,j]^{-1} = E_{-\alpha}[i,j]$
- $$A = \begin{bmatrix} \boxed{1} & 1 & 2 & 0 \\ 3 & 5 & 7 & 1 \\ 1 & 5 & 4 & 5 \\ 2 & 8 & 7 & 9 \end{bmatrix} \xrightarrow{R1} \begin{bmatrix} \boxed{1} & 1 & 2 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix} \xrightarrow{E_{1/2}[2]} \begin{bmatrix} \boxed{1} & 1 & 2 & 0 \\ 0 & \boxed{1} & \frac{1}{2} & \frac{1}{2} \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix}$$
$$\xrightarrow{R2} \begin{bmatrix} \boxed{1} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & \boxed{1} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix} \xrightarrow{E_{1/3}[3]} \begin{bmatrix} \boxed{1} & 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & \boxed{1} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 6 \end{bmatrix} \xrightarrow{R3} \begin{bmatrix} \boxed{1} & 0 & \frac{3}{2} & 0 \\ 0 & \boxed{1} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} = B$$

Here, $R1 = E_{-3}[2,1]$, $E_{-1}[3,1]$, $E_{-2}[4,1]$; $R2 = E_{-1}[2,1]$, $E_{-4}[3,2]$, $E_{-6}[4,2]$; and $R3 = E_{1/2}[1,3]$, $E_{-1/2}[2,3]$, $E_{-6}[4,3]$. The matrix B is the RREF of A. Notice that

$$B = E_{-6}[4,3]\,E_{-1/2}[2,3]\,E_{1/2}[1,3]\,E_{-1/3}[3]\,E_{-6}[4,2]\,E_{-4}[3,2]\,E_{-1}[2,1]E_{-1/2}[2]\,E_{-2}[4,1]\,E_{-1}[3,1]\,E_{-3}[2,1]\,A. = EA$$

$$A = E^{-1}B = E_3[2,1]\,E_1[3,1]\,E_2[4,1]\,E_{-2}[2]\,E_1[2,1]\,E_4[3,2]\,E_6[4,2]$$
$$E_{-3}[3]\,E_{-1/2}[1,3]\,E_{1/2}[2,3]\,E_6[4,3]\,B$$

Suppose $A = [u_1 u_2 u_3 \dots u_n] \in F^{m \times n}$ has RREF B; $B = [Eu_1, Eu_2, \dots, Eu_n]$ be of the form $\theta_i = [a_1\ a_2 \dots a_j\ 0\ 0 \dots 0]^T$

if B has r pivotal columns and has column indexes $k_1 \dots k_r$

$\Rightarrow Eu_{k_1} = e_1$, $Eu_{k_2} = e_2$, $\dots$ $Eu_{k_r} = e_r$

let $\theta_1, \theta_2, \dots \theta_{n-r}$ be non pivotal columns in order $\Rightarrow$

if θ_i occurs blw e_j & e_{j+1}, then θ_i should

$= a_1e_1 + a_2e_2 + \dots a_je_j$

column of A $\quad = a_1Eu_{k_1} + a_2Eu_{k_2} + \dots a_jEu_{k_j}$

$\Rightarrow E^{-1}\theta_i = a_1u_{k_1} + a_2u_{k_2} + \dots + a_ju_{k_j}$

above example: $\theta_1 = \frac{3}{2}u_1 + \frac{1}{2}u_2$

- RREF of a matrix is unique (2 matrices can have same RREF)
- rank(A) = number of pivotal columns in A

- Let $A \in \mathbb{F}^{n\times n}$. The sub-matrix of A obtained by deleting the ith row and the jth column is called the (ij)th minor of A, and is denoted by A_{ij}. The (ij)th co-factor of A is $(-1)^{i+j}\det(A_{ij})$; it is denoted by $C_{ij}(A)$. Sometimes, when the matrix A is fixed in a context, we write $C_{ij}(A)$ as C_{ij}. The adjugate of A is the $n \times n$ matrix obtained by taking transpose of the matrix whose (ij)th entry is $C_{ij}(A)$; it is denoted by $\mathrm{adj}(A)$.

- If A is triangular matrix, det A is product of its diagonal elements.
- $A\,\mathrm{adj}(A) = \mathrm{adj}(A).A = |A|I$
- A square matrix is invertible iff its a product of elemantary matrices (RREF = I)
  We use augmented matrix method here

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 2 & 1 & -1 & -2 & 0 & 0 & 1 & 0 \\ 1 & -2 & 4 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -1 & 2 & 2 & -1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} 1 & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -1 & 2 & 2 & -1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 0 & \boxed{1} & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & -1 & 0 & 1 \end{array}\right]$$

non invertible

$$\left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 2 & 1 & -1 & -2 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & \boxed{-1} & 2 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} \boxed{1} & -1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 3 & -5 & -2 & -2 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & -2 & -2 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & -4 & -2 & -2 & -2 & 0 & 1 \end{array}\right]$$

$$B^{-1} = \frac{1}{7}\begin{bmatrix} 2 & 3 & 4 & 1 \\ 1 & 5 & 2 & -3 \\ 3 & 1 & -1 & -2 \\ 1 & 5 & 2 & \frac{1}{2} \end{bmatrix} \leftarrow \left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & 0 & 2/7 & 3/7 & 4/7 & 1/7 \\ 0 & \boxed{1} & 0 & 0 & 1/7 & 5/7 & 2/7 & -3/7 \\ 0 & 0 & \boxed{1} & 0 & 3/7 & 1/7 & -1/7 & -2/7 \\ 0 & 0 & 0 & \boxed{1} & 1/7 & 5/7 & 2/7 & 1/14 \end{array}\right] \leftarrow \left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & 0 & 6 & 1 & 5 & 2 & 0 \\ 0 & 0 & \boxed{1} & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & \boxed{1} & 1/7 & 5/7 & 2/7 & 1/14 \end{array}\right] \leftarrow \left[\begin{array}{cccc|cccc} \boxed{1} & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & \boxed{1} & 0 & 6 & 1 & 5 & 2 & 0 \\ 0 & 0 & \boxed{1} & 4 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 14 & 2 & 10 & 4 & 1 \end{array}\right]$$

- $A \in \mathbb{F}^{m\times n} \Rightarrow A$ is a function from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$

  $A(x) = Ax \quad x \in \mathbb{F}^{n\times 1}$ — x to Ax

- $A(u+v) = A(u) + A(v)$ $\quad A(\alpha u) = \alpha A(u)$ $\quad \alpha \in \mathbb{F}$
- Range of A $R(A) = \{Ax : x \in \mathbb{F}^{n\times 1}\}$

  $x = [\alpha_1, \alpha_2 .. \alpha_n]^t \Rightarrow R(A) = \{\alpha_1 A_1 + \alpha_2 A_2 + .. \alpha_n A_n\}$

  $A_i \rightarrow i^{th}$ column of A
- vectors $v_1, v_2, .. v_n$ are linearly dependent iff at least one of them can be written as a linear combination of others
- span $\{v_1, v_2, .. v_k\} = \{\alpha_1 v_1 + \alpha_2 v_2 + .. + \alpha_k v_k\}$
- vectors $v_1, v_2, .. v_n$ are linearly independent iff for all $\alpha_1, \alpha_2, .. \alpha_n \in \mathbb{F}$;

  $\alpha_1 v_1 + \alpha_2 v_2 + .. + \alpha_n v_n = 0$ implies $\alpha_1 = \alpha_2 = .. = \alpha_n = 0$
- Superset of linearly dependent set is linearly dependent
- Subset of linearly independent set is linearly independent
- Singleton sets are linearly independent
- For linear independance, we can write:

  (a) vectors as rows and find RREF; if each row has a pivot, then they're linearly independent

(b) vectors as columns and find RREF

- maximum no of linearly independent rows of a matrix = row rank. Similarly column rank. Row rank always = column rank
- Nullity = n - rank(A)
- If $Ax=b$ has a solution (consistent): Rank(A|b) = Rank(A)
- If u is a solution of $Ax=b$, u+y is also a solution, where y is a solution of $Ax=0$
- For a unique solution: rank(A) = n
- For infinitely many solutions: rank(A) < n
- $Ax=b$ has a solution iff $b \in R(A)$
- Gauss Jordan elimination: Take Augment matrix [A|b] and find RREF of A
- A non-empty set V is a subspace of $\mathbb{F}^n$ iff for each $\alpha \in \mathbb{F}$ and for all vectors u, v in V; $\alpha u + v$ must also belong to V

- V is a subspace of F^n and $S \subseteq V \Rightarrow S$ is said to be spanning subset of V / S spans V if $V = \text{span}(S)$
- span(S): set of all linear combinations of finite number of vectors from S
- Let $V \subseteq F^n$ be a subspace; $B \subseteq V$ is the basis of V iff B is linearly independent and span(B) = V. Dimension is number of elements in basis

- $U = \text{span}\{(1, 1, 1, 1), (2, 1, 0, 3), (-1, 0, 1, -2), (0, 3, 2, 1)\}$ Find a basis for the subspace U of $\mathbb{F}^4$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 3 \\ -1 & 0 & 1 & -2 \\ 0 & 3 & 2 & 1 \end{bmatrix} \xrightarrow{R1} \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & -1 & -2 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 3 & 2 & 1 \end{bmatrix} \xrightarrow{R2} \begin{bmatrix} \boxed{1} & 0 & -1 & 2 \\ 0 & \boxed{1} & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & 4 \end{bmatrix} \xrightarrow{R3} \begin{bmatrix} \boxed{1} & 0 & 0 & 1 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow [1,1,1,1], [2,1,0,3], [0,3,2,1]$

The pivotal rows show that $\{(1, 0, 0, 1), (0, 1, 0, 1), (0, 0, 1, -1)\}$ is a basis for the given subspace

- Range of linear transformation $R(A) = \{Ax : x \in F\}$
  $R(A) = \text{span}\{\text{columns of } A\}$
- Null space: $x : \{Ax = 0\}$
- $\dim(R(A))$ [rank] + $\dim(N(A))$ [nullity] = n

$$A = \begin{bmatrix} 5 & 2 & -3 & 1 \\ 1 & -3 & 2 & -2 \\ 3 & 8 & -7 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} \boxed{1} & 0 & -5/17 & -1/17 \\ 0 & \boxed{1} & -13/17 & 11/17 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \text{RREF}(A)$$

a basis for $R(A)$ is $\{[5, 1, 3]^t, [2, -3, 8]^t\}$.

For a basis of $N(A)$, notice that each pivot has the row index equal to the column index; so, we do not require to insert zero rows between pivotal rows. To make it a square matrix, we attach a zero row to the RREF at the bottom:

$$D = \begin{bmatrix} \boxed{1} & 0 & -5/17 & -1/17 \\ 0 & \boxed{1} & -13/17 & 11/17 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Then, we change the diagonal entries in the non-pivotal columns to -1. These changed non-pivotal columns form a basis for $N(A)$. That is,

a basis for $N(A)$ is $\{[-\frac{5}{17}, -\frac{13}{17}, -1, 0]^t, [-\frac{1}{17}, \frac{11}{17}, 0, -1]^t\}$.

- let $B=\{v_1, v_2 .. v_m\}$ be an ordered set of vectors from subspace V. B is a basis of V iff for each $v \in V$, there exists a unique vector $[\alpha_1, \alpha_2 .. \alpha_m]^t$ such that $v = \alpha_1 v_1 + \alpha_2 v_2 + .. \alpha_m v_m$. This unique vector is called the coordinate vector of v w.r.t basis B i.e $[v]_B$.

- **Theorem 3.8** *Let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{F}^{n\times 1}$, and let $B = \{u_1, \ldots, u_n\}$ be any basis of $\mathbb{F}^{n\times 1}$. Write $P = [u_1 \cdots u_n]$. Then, $P[e_j]_B = e_j = [u_j]_B$ for $1 \le j \le n$, and for each $u \in \mathbb{F}^{n\times 1}$, $P[u]_B = u$ and $[u]_B = P^{-1}u$.*

Proof Let $j \in \{1, \ldots, n\}$. By Observation 3.2, $[u_j]_B = e_j$. Since u_j is the jth column of P, we have $u_j = Pe_j = P[u_j]_B$.

Let $u \in \mathbb{F}^{n\times 1}$. Then, $u = a_1u_1 + \cdots + a_nu_n$ for unique scalars $a_1, \ldots, a_n$. Now,

$$\begin{aligned} P[u]_B &= P\left[a_1, \ldots, a_n\right]^t = P(a_1e_1 + \cdots + a_ne_n) \\ &= a_1Pe_1 + \cdots + a_nPe_n = a_1u_1 + \cdots + a_nu_n = u. \end{aligned}$$

As the columns of P form a basis for $\mathbb{F}^{n\times 1}$, the matrix P is invertible. Therefore, $[u]_B = P^{-1}u$. ■

- $A \in \mathbb{F}^{m\times n} \Rightarrow$ A is linear transformation from $\mathbb{F}^{n\times 1}$ to $\mathbb{F}^{m\times 1}$. let D, E be standard bases of $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$ respectively, then $[u]_D = u$ and $[Au]_E = A[u]_D$
- let $A \in \mathbb{F}^{m\times n}$; $B=\{u_i\}$ and $C=\{v_j\}$ are bases of $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$ respectively. There exists a unique

matrix M such that $[Au]_C = M[u]_B$ $\quad M = [A]_{C,B}$

- Compute the coordinate matrix $[A]_{C,B}$ of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$ with respect to the bases $B = \{[1,2]^t, [3,1]^t\}$ for $\mathbb{F}^{2\times 1}$, and $C = \{[1,0,0]^t, [1,1,0]^t, [1,1,1]^t\}$ for $\mathbb{F}^{3\times 1}$.

Write $u_1 = [1,2]^t, u_2 = [3,1]^t$ and $v_1 = [1,0,0]^t, v_2 = [1,1,0]^t, v_3 = [1,1,1]^t$. We obtain

$$\Rightarrow \begin{aligned} Au_1 &= [2,3,-1]^t &&= -v_1 + 4v_2 - v_3 \\ Au_2 &= [1,4,2]^t &&= -3v_1 + 2v_2 + 2v_3. \end{aligned} \Rightarrow [A]_{C,B} = \begin{bmatrix} -1 & -3 \\ 4 & 2 \\ -1 & 2 \end{bmatrix}.$$

- $$\begin{aligned} Au_1 &= \alpha_{11}v_1 + \alpha_{21}v_2 + \cdots + \alpha_{m1}v_m \\ Au_2 &= \alpha_{12}v_1 + \alpha_{22}v_2 + \cdots + \alpha_{m2}v_m \\ &\vdots \\ Au_n &= \alpha_{1n}v_1 + \alpha_{2n}v_2 + \cdots + \alpha_{mn}v_m \end{aligned} \tag{3.1}$$

Then, the coordinate matrix is (Mark which coefficient α_{ij} goes where)

$$[A]_{C,B} = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ & \vdots & & \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{bmatrix}. \tag{3.2}$$

- equivalent: $B = Q^{-1}AP$
- similar: $B = P^{-1}AP$
- Any matrix can be written as product of full-rank matrices

- **Theorem 3.10** *Let $A \in \mathbb{F}^{m\times n}$. Let $B = \{u_1, \ldots, u_n\}$ and $C = \{v_1, \ldots, v_m\}$ be bas for $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively. Construct the matrices*

$$P = [u_1 \quad \cdots \quad u_n], \quad Q = [v_1 \quad \cdots \quad v_m].$$

by taking the column vectors u_j and v_i as columns of the respective matrices. The $[A]_{C,B} = Q^{-1}AP.$

- For change of basis vector $[I]_{N,O}$, just put $A = I$ above, N is new basis i.e Q and O is old basis i.e P $\Rightarrow I_{N,O} = N^{-1}O$
- inner product $\langle x\, y\rangle = y^* x$

$u = (a_1 \ldots a_n)$, $v = (b_1, \ldots b_n) \Rightarrow \langle u, v\rangle = \sum a_i \overline{b_i}$

- Norm $\|x\| = (\langle x\, x\rangle)^{1/2}$ $\quad$ - $|\langle x, y\rangle| \le \|x\|\,\|y\|$

- *Gram–Schmidt Orthonormalization*:

$$\begin{aligned} v_1 &= u_1; & w_1 &= v_1/\|v_1\| \\ v_2 &= u_2 - \langle u_2, w_1\rangle w_1; & w_2 &= v_2/\|v_2\| \\ &\vdots \\ v_m &= u_m - \langle u_m, w_1\rangle w_1 - \cdots - \langle u_m, w_{m-1}\rangle w_{m-1}; & w_m &= v_m/\|v_m\|. \end{aligned}$$

$z = x - \frac{\langle x\, y\rangle}{\langle y\, y\rangle} y \quad \langle z, z\rangle > 0$

- $\|x+y\| \le \|x\| + \|y\|$
- orthogonal: $\langle x, y\rangle = 0$

Theorem 4.2 (Gram–Schmidt orthogonalization) *Let $u_1, \dots, u_m \in \mathbb{F}^n$. Let $1 \le k \le m$. Define*

$$
\begin{aligned}
v_1 &= u_1 \\
v_2 &= u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 \\
&\vdots \\
v_k &= u_k - \frac{\langle u_k, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \cdots - \frac{\langle u_k, v_{k-1} \rangle}{\langle v_{k-1}, v_{k-1} \rangle} v_{k-1}.
\end{aligned}
$$

Then the following are true:

(1) $\operatorname{span}\{v_1, \dots, v_k\} = \operatorname{span}\{u_1, \dots, u_k\}$.
(2) If $\{u_1, \dots, u_k\}$ is linearly independent, then $\{v_1, \dots, v_k\}$ is orthogonal.
(3) If $u_k \in \operatorname{span}\{u_1, \dots, u_{k-1}\}$, then $v_k = 0$.

Theorem 4.3 (QR-factorization) *Let the columns of $A \in \mathbb{F}^{m \times n}$ be linearly independent. Then there exist a matrix $Q \in \mathbb{F}^{m \times n}$ with orthonormal columns, and an upper triangular invertible matrix $R \in \mathbb{F}^{n \times n}$ such that $A = QR$. Further, if the diagonal entries in R are real, and they are either all positive, or all negative, then both Q and R are uniquely determined from A.*

Proof Write the columns of $A \in \mathbb{F}^{m \times n}$ as $u_1, u_2, \dots, u_n$ in that order. These vectors are linearly independent in $\mathbb{F}^{m \times 1}$. So, $m \ge n$. Using Gram–Schmidt orthonormalization on the list of vectors $u_1, \dots, u_n$, we obtain an orthonormal list of vectors, say, $w_1, w_2, \dots, w_n$. Now, $\operatorname{span}\{u_1, \dots, u_k\} = \operatorname{span}\{w_1, \dots, w_k\}$ for $1 \le k \le n$. In particular, u_k is a linear combination of $w_1, \dots, w_k$.

Hence there exist scalars a_{ij}, $1 \le i \le j \le n$ such that

$$
\begin{aligned}
u_1 &= a_{11} w_1 \\
u_2 &= a_{12} w_1 + a_{22} w_2 \\
&\vdots \\
u_k &= a_{1k} w_1 + \cdots + a_{kk} w_k \\
&\vdots \\
u_n &= a_{1n} w_1 + \cdots + a_{kn} w_k + \cdots + a_{nn} w_n
\end{aligned}
$$

We take $a_{ij} = 0$ for $i > j$, and form the matrix

$$
R = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{bmatrix}.
$$

Theorem 4.4 *Let $\{v_1, \dots, v_m\}$ be an orthonormal basis for a subspace V of $\mathbb{F}^n$. Then the following are true for each $x \in V$:*

(1) **(Fourier Expansion)** $x = \langle x, v_1 \rangle v_1 + \cdots + \langle x, v_m \rangle v_m$.
(2) **(Parseval Identity)** $\|x\|^2 = |\langle x, v_1 \rangle|^2 + \cdots + |\langle x, v_m \rangle|^2$.

best approximation
||
$\text{proj}_V(x)$

Theorem 4.5 *Let V be a subspace of $\mathbb{F}^n$. Corresponding to each $x \in \mathbb{F}^n$, there exists a unique vector $y \in V$ such that $x - y \perp V$. Further, if $\{v_1, \ldots, v_m\}$ is any orthonormal basis of V, then such a vector y is given by $y = \sum_{j=1}^{m} \langle x, v_j \rangle v_j$.*

For this, let $\{v_1, \ldots, v_m\}$ be an orthonormal basis of a subspace V of $\mathbb{F}^{n\times 1}$. Let $x \in \mathbb{F}^{n\times 1}$. Write $z = [c_1, \cdots, c_m]^t \in \mathbb{F}^{m\times 1}$ with $c_j = \langle x, v_j \rangle = v_j^* x$, and $P = [v_1 \quad \cdots \quad v_m]$. Then

$$z = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} v_1^* x \\ \vdots \\ v_m^* x \end{bmatrix} = P^* x, \quad y = \sum_{j=1}^{m} \langle x, v_j \rangle v_j = \sum_{j=1}^{m} c_j v_j = Pz = PP^* x.$$

Notice that $PP^* x = \text{proj}_V(x)$ for each $x \in \mathbb{F}^{n\times 1}$. Due to this reason, the matrix $PP^* \in \mathbb{F}^{n\times n}$ is called the **projection matrix** that projects vectors of $\mathbb{F}^{n\times 1}$ onto the subspace V.

Example 4.5 Let $V = \text{span}\{[1, 0, -1]^t, [1, 1, 1]^t\}$ and let $x = [1, 2, 3]^t$. Compute the projection matrix that projects vectors of $\mathbb{F}^{3\times 1}$ onto V, and $\text{proj}_V(x)$.

Since the vectors $[1, 0, -1]^t$ and $[1, 1, 1]^t$ are orthogonal, an orthonormal basis for V is given by $\{v_1, v_2\}$, where

$$v_1 = [1/\sqrt{2},\ 0,\ -1/\sqrt{2}]^t, \quad v_2 = [1/\sqrt{3},\ 1/\sqrt{3},\ 1/\sqrt{3}]^t.$$

With $P = [v_1 \quad v_2]$, the projection matrix is given by

$$PP^* = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} = \begin{bmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & 5/6 \end{bmatrix}.$$

Then $\text{proj}_V(x) = PP^* x = [1, 2, 3]^t$.

Also, directly from the orthonormal basis for V, we obtain

$$\text{proj}_V(x) = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 = [1, 2, 3]^t.$$

Indeed, $x = -1[1, 0, -1]^t + 2[1, 1, 1]^t \in V$. Thus, $\text{proj}_V(x) = x$. □

Theorem 4.8 *Let $A \in \mathbb{F}^{m\times n}$ and let $b \in \mathbb{F}^{m\times 1}$.*

(1) The linear system $Ax = b$ has a least squares solution.
(2) A vector $u \in \mathbb{F}^{n\times 1}$ is a least squares solution of $Ax = b$ iff $Au - b \perp v$ for each $v \in R(A)$.
(3) A vector $u \in \mathbb{F}^{n\times 1}$ is a least squares solution of $Ax = b$ iff it is a solution of $A^*Ax = A^*b$.
(4) A least squares solution is unique iff $N(A) = \{0\}$.

Theorem 4.9 *Let $A \in \mathbb{F}^{m\times n}$ have linearly independent columns. Let $A = QR$ be a QR-factorization of A. Then, the least squares solution of $Ax = b$ is given by $u = R^{-1}Q^*b$.*

- characteristic polynomial : $|\lambda I - A| = 0$

$\lambda \rightarrow$ eigenvalue which satisfies $Av = \lambda v$,

v is corresponding eigenvector

- eigenvectors corresponding to distinct eigenvalues are

linearly independent

- spectrum: set of eigenvalues of a matrix (including repititions)
- $\sigma(A) = \{\lambda_1, \dots \lambda_n\} \Rightarrow tr(A) = \sum \lambda_i$, $det(A) = \prod \lambda_i$
- Hermitian: $A^* = A$ skew hermitian: $A^* = -A$

  unitary: $AA^* = A^*A = I$ normal: $AA^* = A^*A$ (orthogonal for real)

$(\lambda - \mu)\langle x, y\rangle = \lambda\langle x, y\rangle - \mu\langle x, y\rangle$
$= \langle \lambda x, y\rangle - \langle x, \mu y\rangle$
$= \langle Ax, y\rangle - \langle x, Ay\rangle$
$= 0 \Rightarrow \langle x, y\rangle = 0$

- **Theorem 5.5** *Let $A \in \mathbb{R}^{n \times n}$ and let λ be any eigenvalue of A.*
  1. *If A is hermitian, then $\lambda \in \mathbb{R}$; and eigenvectors corresponding to distinct eigenvalues of A are orthogonal.*
  2. *If A is real symmetric, then $\lambda \in \mathbb{R}$; and there exists a real eigenvector corresponding to λ.*
  3. *If A is skew hermitian or real skew symmetric, then λ is purely imaginary or zero.*
  4. *If A is unitary or orthogonal, then $|\lambda| = 1$.*

$Ax = \lambda x \Rightarrow x^*A^* = \bar{\lambda} x^* \Rightarrow x^*A = \bar{\lambda} x^*$
$\Rightarrow x^*Ax = \bar{\lambda} x^*x \quad x^* \cdot \lambda x = \bar{\lambda} x^*x \Rightarrow \lambda = \bar{\lambda}$

$Ax = \lambda x \Rightarrow x^*A^* = \bar{\lambda} x^* \Rightarrow -x^*A = \bar{\lambda} x^*$
$\Rightarrow -x^*Ax = \bar{\lambda} x^*x \Rightarrow -x^*\lambda x = \bar{\lambda} x^*x \Rightarrow \lambda = -\bar{\lambda}$

$Ax = \lambda x \Rightarrow x^*A^* = \bar{\lambda} x^* \Rightarrow x^*A^*Ax = \bar{\lambda} x^* \cdot \lambda x$
$\Rightarrow x^*x = |\lambda|^2 x^*x \rightarrow |\lambda| = 1$

- Schur's triangularization:

  $A = PBP^*$ $P \rightarrow$ unitary $B \rightarrow$ upper Δ

  Find eigenvector v_1 of A corresponding to eigenvalue λ_1. Extend the eigenvector to an ONB and let P_1 be matrix with those as columns. Find $P_1^*AP_1$. Now apply the same for the resulting matrix but excluding row 1 and column 1.

  $P = P_1 \begin{bmatrix} 1 & 0 \\ 0 & P_2 \end{bmatrix}$ and $B = P^*AP$

- $P(A) = c_0 I + c_1 A + \dots + c_k A^k \Rightarrow P(t)$ annihilates A if $P(A) = 0^n$. $\chi_A(A)$ annihilates A (Cayley Hamilton)

  $m_A(t)$ (minimal polynomal) - polynomial with min degree that annhilates a matrix.

  $\chi_A(t) = q(t) \cdot m_A(t)$.

- If A is diagonal matrix and $\chi_A(t) = (t-\lambda_1)^{m_1} (t-\lambda_2)^{m_2} \dots \Rightarrow m_A(t) = (t-\lambda_1)(t-\lambda_2)\dots$

- Diagonalization:

  $A = PDP^{-1}$ P is invertible

  $\sigma(A) = \sigma(D) = \{d_1, d_2, \dots d_n\}$

  $P = [u_1\ u_2 \dots u_n] \Rightarrow u_i$ is eigenvector of A with eigenvalue d_i

  If A is diagonalizable:

  (i) there exists a basis consisting of eigenvectors of A

  (ii) geometric multiplicity $\dim(N(A-\lambda I))$ of each eigenvalue of A is equal to its algebraic multiplicity $(GM \le AM)$

  $AM \Rightarrow (t-\lambda)^k : k \Rightarrow AM$

(iii) Sum of geometric multiplicities of all eigenvalues of A is n

(iv) $m_A(t) = (t-\lambda_1)\dots(t-\lambda_k)$; $\lambda_1, \dots \lambda_k \to$ scalars

- Singular value Decomposition

$\sigma(A^*A)$ or $\sigma(AA^*) = \{\lambda_1, \dots \lambda_n\}$.

For each $\lambda_i > 0$; $s_i = \sqrt{\lambda_i}$ (singular value)

rank(A) = no of the singular values of A

let $A \in C^{m \times n}$; rank $= r$; $s_1, \dots s_r \to$ singular values

$$\Sigma = \begin{bmatrix} S & 0_{m \times (n-r)} \\ 0_{(m-r) \times n} & 0_{(m-r) \times (n-r)} \end{bmatrix}_{m \times n} \quad S = \text{diag}(s_1, s_2, \dots s_r)$$

$s_1 > s_2 > \dots s_r$ (preferred)

$U = [u_1 \dots u_m] \in C^{m \times m}$; $V = [v_1 \dots v_n] \in C^{n \times n}$

$U, V \to$ unitary

(i) $A = U \Sigma V^* = s_1 u_1 v_1^* + s_2 u_2 v_2^* + \dots s_r u_r v_r^*$

(ii) $AA^* u_i = \begin{cases} s_i^2 u_i & 1 \le i \le r \\ 0 & r < i < m \end{cases}$ $\quad A^*A v_i = \begin{cases} s_i^2 v_j & 1 \le j \le r \\ 0 & r < j < n \end{cases}$

(iii) $u_i = \frac{Av_i}{s_i}$; $v_i = \frac{A^* u_i}{s_i}$ $\quad \forall\ 1 \le i \le r$

(iv) $\{u_1, \dots u_r\} \to$ ONB for R(A)

$\{v_1, \dots v_r\} \to$ ONB for $R(A^*)$

$\{u_{r+1}, \dots u_m\} \to$ ONB for $N(A^*)$

$\{v_{r+1} \dots v_n\} \to$ ONB for N(A)

- Thin SVD: $A \in C^{m \times n}$ has rank r ; $K = \min(m,n)$

  $A = \bar{U} \bar{\Sigma} \bar{V}^{*}$ $\quad \bar{\Sigma} = \text{diag}(S_1, S_2 \dots S_r, 0, \dots 0) \in C^{K \times K}$

  $\bar{U} \in C^{m \times K}$, $\bar{V} \in C^{n \times K}$ have orthonormal columns

- Tight SVD: $S = \text{diag}(S_1, \dots S_r)$. First r columns of u and $V = \hat{u}, \hat{V} \Rightarrow A = \hat{u} S \hat{V}^{*}$

- Jordan form: improvized SVD

$J_n(\lambda)$: Jordan block = diagonal elements $= \lambda$
super diagonal elements $= 1$
others $= 0$

A matrix A is said to be in Jordan form if $A = \text{diag}(J_{n_1}(\lambda_1), J_{n_2}(\lambda_2) \dots J_{n_K}(\lambda_K))$

$A = PJP^{T}$

$m_K(\lambda)$: no of Jordan blocks of order K, diagonal element λ in J $= \text{rank}((A-\lambda I)^{K-1}) + \text{rank}((A-\lambda I)^{K+1}) - 2 \cdot \text{rank}((A-\lambda I)^{K})$

Diagonal entries of J are eigenvalues of A

Number of Jordan block $= GM(\lambda) = \dim(N(A-\lambda I))$ with diagonal entries $\lambda \in \sigma(A)$

$P = [v_1 \dots v_n] \Rightarrow v_1 \in N(A-\lambda I) \dots v_K \in N((A-\lambda I)^{K})$